

Incident Runbook Internal Learning Management System

Severity	SEV 2 by default • Raise to SEV 1 for security, data-integrity, or broad service impact
Critical flow	Course assignment → quiz attempt → completion → certificate generation and email

Alarm meaning Error-rate alarm: more than 5% Lambda errors for two consecutive 5-minute periods. Latency alarm: Lambda P95 duration above 3,000 ms for two consecutive 5-minute periods.

POSSIBLE CAUSES

- 401: missing, invalid, or expired Cognito JWT; 403: learner/admin group mismatch
- Quiz/course data mismatch or attempts-limit validation failure
- DynamoDB AccessDenied on completion write, conflict, or throttling
- S3 certificate encryption/PutObject permission error
- PDF timeout, oversized content, or SES delay/rejection

BUSINESS IMPACT

Learners may be unable to submit quizzes, completion status may not save, certificates may be missing, or certificate emails may be delayed.

FIRST RESPONSE

1. Acknowledge the alert and record alarm, time, region, course ID, and correlation ID.
2. For 401/403, validate JWT expiry/issuer/audience, group claim, Cognito authorizer, and route protection; GET /verify/{cert_id} is the intentional public exception.
3. Determine whether failure is in grading, completion write, PDF generation, S3 storage, or SES delivery using logs and X-Ray.
4. For AccessDenied, match the denied DynamoDB/S3 action to the exact resource ARN; never restore wildcard permissions.
5. Retry certificate generation idempotently only after confirming no valid certificate already exists. Never consume another quiz attempt during recovery.

RECOVERY VERIFICATION

- Run a test quiz and confirm one completion record is stored.
- Open the generated PDF from the expected S3 key and validate the public verification endpoint if used.
- Confirm email status and that alarms remain OK for two periods.

ESCALATION CONDITIONS

- Failures continue for 15 minutes or affect multiple courses/users.
- Quiz attempts or completion records are inconsistent, lost, or duplicated.
- Certificates expose incorrect learner data, S3 access is public, or AWS dependency degradation is suspected.

Recovery evidence to attach Alarm graph; correlation ID; quiz/completion IDs; Lambda logs; S3 object key; SES event; test result.